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New Role of Technical Specialists to Enable Digital Transformation in the Petroleum Industry: A Petrophysicist-Based Proof of Concept

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Standard petrophysical workflows are limited by centralized control of log data, restricting tool access to dedicated specialists. This structure often leads to delays, reduced reproducibility and limited accessibility for multidisciplinary teams. To address these challenges, this study proposes a methodological framework combining natural language interfaces with an GenAI-based orchestration layer. The approach allows non-petrophysical specialists to access key petrophysical data, perform standard analyses, reduce reliance on manual processing and ensure secure, reproducible operations.

The system is built on a modular architecture to streamline and standardize petrophysical data analysis. The Natural Language Interface uses large language models to interpret user queries and translate them into structured tool commands. The MCP Server Execution layer orchestrates analytical workflows, manages access control, and coordinates the execution of core analytical tools. Library Operations provide the fundamental well log processing, quality control, statistical analysis, and data visualization. The containerized system supports local deployment, ensuring traceability and data confidentiality within secure corporate infrastructure.

Results indicate that the proposed agent-based framework mitigates procedural constraints associated with centralized log interpretation, facilitating more autonomous access for subsurface teams. The natural language interface translates user queries into structured analytical sequences, supporting reproducible execution and preserving auditability. This outcome supports statistical summaries, anomaly indicators, and customized visualizations optimized for interpretation by multidisciplinary teams.

The framework enables petrophysicists to transition from task execution to system-level oversight, focusing on the design, validation, and supervision of automated workflows.

Data security is a critical concern, addressed through containerized local deployment and the use of private LLM instances, ensuring compliance with internal data governance policies and eliminating external risks. The system integrates internal corporate data with locally hosted open-source libraries deployed under strict network isolation to ensure that no data is transmitted outside the organizational environment.

The results confirm the feasibility of the approach as a scalable solution for restructuring interpretation workflows and reducing barriers in geoscientific collaboration.

This study introduces a secure framework for embedding natural language-driven AI agents into subsurface workflows. A key impact is the shift in the petrophysicist's role from repetitive execution to workflow supervision and agent training. Significant business impact by reducing workload and decisions acceleration while supporting full data control and business process modularity. The cognitive layer uses structured prompting, including chain-of-thought reasoning to enhance interpretability and robustness of responses.

Introduction

Digital technologies and artificial intelligence are transforming mostly focusing only on some elements of subsurface workflows in the energy industry (Michael, Srivastava, Garg, & Harkouss, 2020) (Thambynayagam, 2025) (Tveritnev, et al., 2023). Despite progress in data acquisition, storage, and processing, the practical value of these datasets often remains often remains under-utilised due to conservative organizational practices and limited access restricted to a small group of domain experts (Heine et al., 2023). In most companies, petrophysical interpretation continues to be a highly centralized and routine process, with both data ownership and interpretation capabilities limited to a small number of experts. This often leads to data exchange delays and extends project timelines.

This study solves these critical challenges proposing a novel modular architecture. Suggested architecture designed to optimize data accessibility and increasing efficiency of company processes, and presents not only a conceptual research framework but also a fully functional proof-of-concept implementation, demonstrating applicability and practical benefits for industry implementation.

Objectives and Scope

The primary objective of this study is to implement and validate a modular agent-based digital workflow, integrating natural language interfaces with a server based on Model Context Protocol, MCP (Anthropic, 2025). Proposed in this study modular architecture focuses on the improvement of multidisciplinary access, data reproducibility, and transparency in petrophysical processes. The main goal is a digital transformation in subsurface workflow and changing the role of the petrophysicist, who now acts as a hybrid expert and developer, responsible for designing and verifying the correctness of MCP protocol and servers and implement new analytical modules for the generative AI agent, which further performs part of the routine work for the specialist. This allows the petrophysicist to spend more time on challenging analysis and tasks and consulting.

The scope is limited to internal integration within corporate IT environments. This clear boundary aligns the proposed system with stringent corporate data governance and security requirements.

Methodology

Conceptual Framework Overview

This study introduces a modular, generative artificial intelligence (GenAI) agent digital architecture that is specifically engineered to allow non-specialists access to petrophysical information and automate interpretation workflows that have previously been performed manually.

The proposed framework is based on a layered architecture (Fig. 1). Each layer is responsible for a specific part of the workflow.

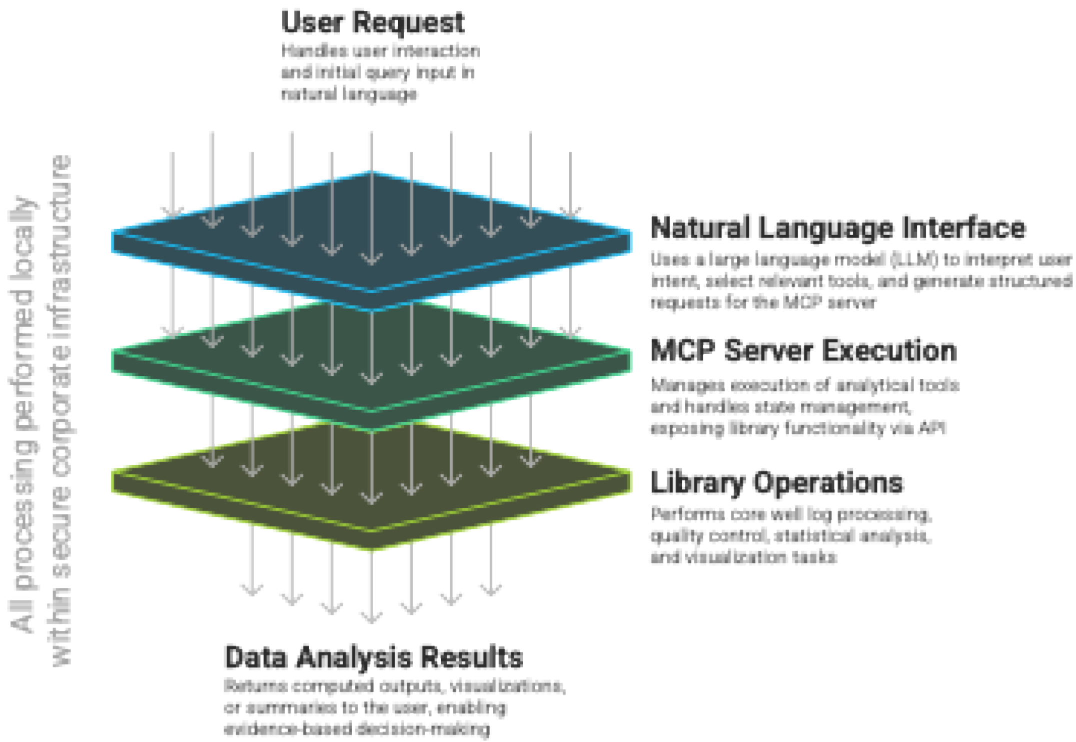

Figure 1—Modular agent based Architecture for digital petrophysical workflow.

This modular framework uses artificial intelligence and natural language processing (NLP) technologies, allowing users to interact intuitively with subsurface data, despite their coding expertise or experience with specialized petrophysical software. The system reduces the technical barriers associated with traditional petrophysical data analysis workflows by removing analytical complexity using natural language requests and generative AI driven orchestration.

System Architecture and Design

User Request. The process begins when a user submits a request or task via a simple chat-based user interface using natural language of choice. This approach enables technical specialists to initiate complex analytical routines without needing to understand underlying data structures or coding expertise.

Natural Language Interface. The system utilizes state-of-the-art transformer-based large language models, LLMs (Vaswani, et al., 2017) (NVIDIA, 2025) (Radford, Narasimhan, Salimans, & Sutskever, 2018) (Radford, Wu, Child, & Sutskever, 2019), specifically in the current configuration Gemini 2.5 Pro model (Google DeepMind, 2025). This model was selected due to its long context window and strong reasoning performance, which are well-suited for interpreting petrophysical data and workflows. The model does not contain an explicit knowledge base, instead it encodes parametric knowledge in its weights learned from large text corpora. Therefore, factual completeness and freshness are not guaranteed without external tools. For up-to-date or proprietary facts, the system integrates retrieval/tool-use to ground responses in verified sources. In internal evaluations, Gemini 2.5 Pro produced the most consistent, logically coherent, and context-aware outputs when instructed to act as a petrophysicist, outperforming the other models we tested under identical prompts and datasets.

This layer enables

- Natural-language access that lowers barriers for non-petrophysicists (e.g., geologists, reservoir engineers, and others).

- Structured prompting (Liu, et al., 2023) (Chen, Zhang, Langrené, & Zhu, 2025) and multi-step reasoning (Wei, et al., 2023) (Wang, et al., 2023) where prompt templates decompose tasks into structured, executable MCP commands with intermediate steps are made transparent via logs and result artifacts. This provides logical, transparent, and auditable results.
- Workflow orchestration including sequential or iterative analytical workflows are coordinated automatically, enabling multi-step analyses without manual scripting or direct user orchestration.

MCP Server Execution. The proposed Model Context Protocol server (Ermilov, Tveritnev, & Trusova, AlmazErmilov/welly-fork: Welly helps with well loading, wireline logs, log quality, data science, 2025) provides a standardized interface in the integration of digital workflows for petrophysical analysis. Historically, each analytical tool or software in a petrophysicist's toolkit required a custom integration directly. As the number of software and custom petrophysical workflows and codes increases, it becomes difficult to operate the processes manually and to transfer knowledge and best practices within the company.

A centralized MCP server allows abstraction of analytical tools and libraries behind a single, standardized API. In this work, we implement the MCP server for petrophysical tasks so that, instead of maintaining one-off interactions with each tool, all orchestration and data exchange flow through the MCP protocol.

With large language models processing natural-language requests, the MCP server acts as a bridge between user intent and computation.

- It receives structured, context-rich requests generated by the LLM in response to user queries.
- It manages session state and coordinates the sequential execution of required analytical routines.
- It invokes specialized functions from dedicated petrophysical libraries and software.

As a result, new analytical workflows can be orchestrated rapidly and reliably, regardless of the underlying complexity or the user's technical background.

The centralized model offers several practical benefits.

- Simplified integration provides access to tools through a single protocol and a shared session context, which reduces ad hoc integrations and improves maintainability.
- Transparent API documentation is produced automatically because the MCP server generates Swagger pages that describe endpoints consistently.
- Faster innovation follows from the ability to add or update analytical tools with minimal changes to the overall system, which shortens R&D and productization cycles.
- LLM-based orchestration connects natural-language interfaces with specialized computational tools which enables non-specialists to run standard analyses.

Where programmatic access is available, the LLM can dispatch structured requests across multiple MCP servers and adapters for proprietary applications. This design allows integration with platforms such as Petrel (SLB, 2025) and Techlog (SLB, 2025) as well as in-house tools and libraries, while keeping user interaction at a single entry point. The orchestration layer selects the target tool, maintains session context, enforces access control, and records the execution.

Previously, each analytical request required direct communication with a petrophysicist, who would search for a solution using specialized software or custom code. Every new tool or method demanded separate integration and often relied on individual expertise, which made the process labor-intensive and hard to scale. Fig. 2 illustrates the traditional approach to petrophysical data analysis.

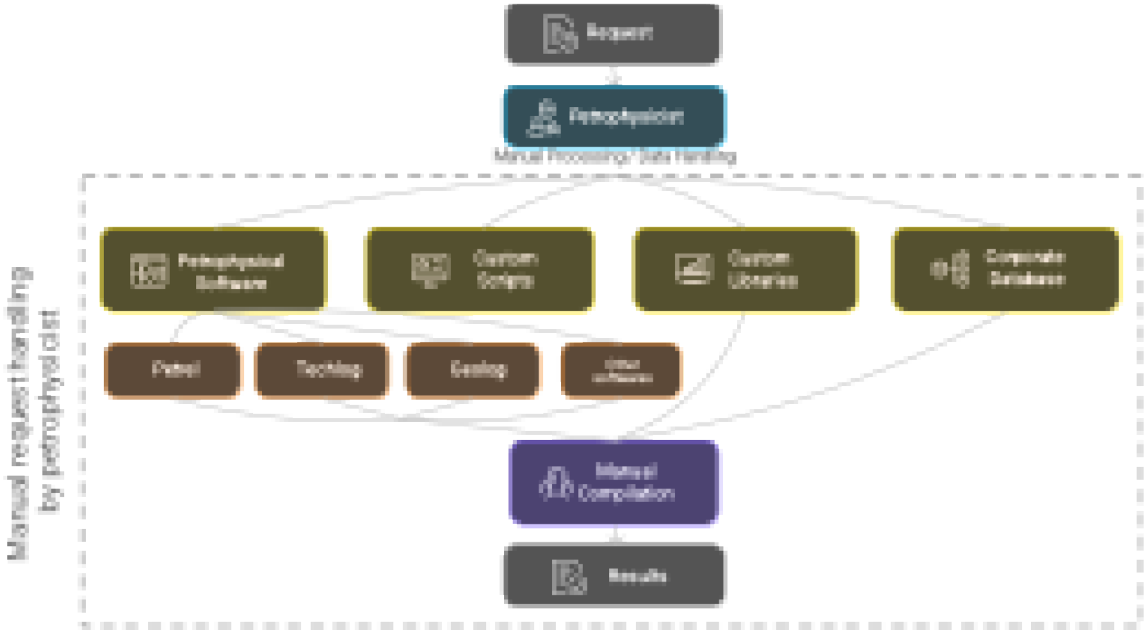

Figure 2—Traditional manual workflow in petrophysical data analysis.

Now, with the MCP approach, users submit questions in plain language. An orchestration controller implemented via the MCP interface receives each request, selects the appropriate software tool, library, or analytical module, and organizes the computation. The petrophysicist's role shifts to configuring and extending the MCP server, integrating new analytical libraries, and validating module behavior, which reduces routine work. This approach shown in Fig. 4 democratizes access to complex analysis and accelerates subsurface workflows.

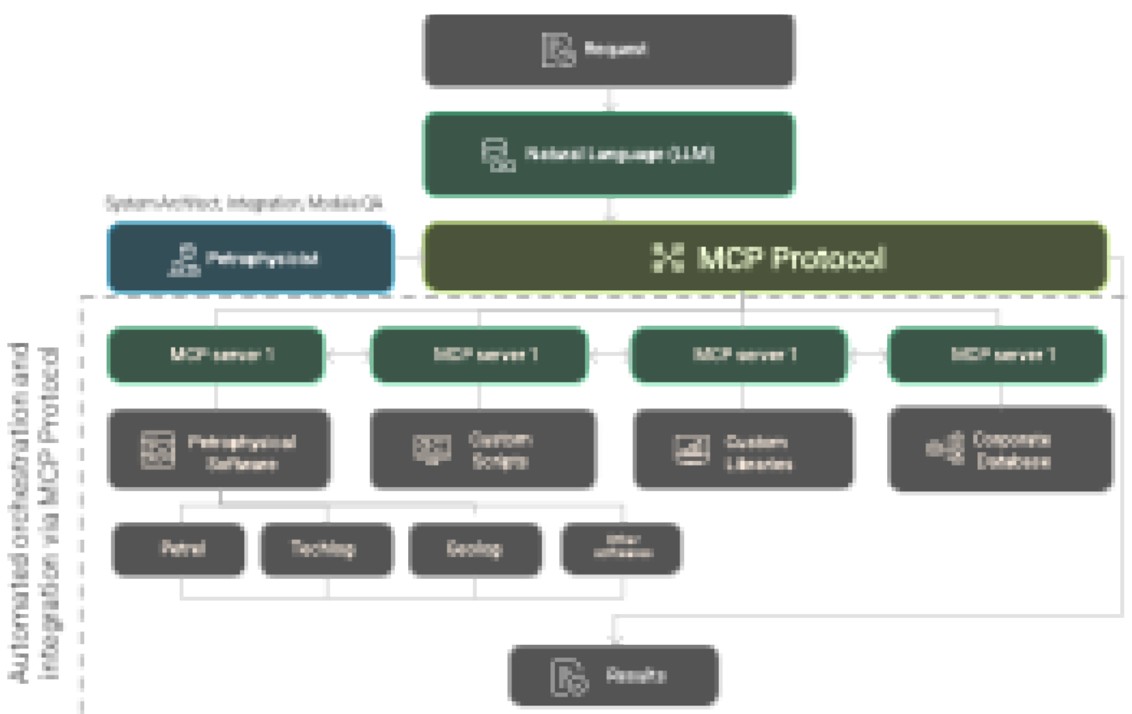

Figure 4—Automated digital workflow with MCP protocol in petrophysical analysis.

Library Operations. Specialized software libraries perform the core computations including well log processing, statistical data quality control, statistical analysis, and generation of visualizations. For LAS file (Canadian Well Logging Society, 2017) processing and well-log analytics, the system uses open-source tools designed specifically for the petrophysics tasks (Agile*, 2025) (Inverarity, 2025).

Core analytical tools implemented in the library are LAS file loading and validation, curve statistics extraction, well log plotting and visualization, well metadata extraction, curve enumeration and batch statistics.

Data Analysis Results. The system returns computed outputs such as statistical summaries and plots in real time. The frontend shows the results as interactive charts and short summaries that help teams review evidence and make decisions together.

Security and Data Governance Considerations

Because subsurface data are sensitive and often proprietary, the system includes strict data-governance measures.

- Containerized deployment packages all components with Docker (Docker, 2025) and Docker Compose (Docker Inc., 2025), teams can reproduce runs, track changes, and maintain the system with less effort.
- Local or private hosting keeps the full workflow inside the company network. Strict network isolation prevents data from leaving environment.
- Logging and audit record every operation, data access, and error. The logs support compliance checks and internal quality reviews.
- LLM security options let teams use local open-source models to keep data on premises or use managed proprietary models (for example, OpenAI models via Microsoft Azure) with enterprise controls such as private networking and key management.

These measures increase accessibility and transparency while protecting the confidentiality of subsurface data.

Implementation

Technical Implementation Overview

The implementation of the proposed digital petrophysical workflow is accomplished through an integrated technological stack combining Python-based analytical backends, an LLM driven cognitive orchestration layer and a frontend interface. The goal is a secure and easy to use system that lets a user to work with petrophysical data in natural language, no matter their programming skills.

The key technical components and their integrations include

- **LLM and NLP Orchestration** powered by Gemini 2.5 Pro in our case to understand natural-language requests, create structured API calls, and run multi-step analyses.
- **Backend MCP Server** implemented using FastAPI in Python (Ramírez, 2025), incorporating the open-source library (Agile*, 2025) for petrophysical data manipulation and matplotlib (Hunter, 2007) for data visualization, and can be plugged in into various chat interfaces.

Frontend Implementation

The frontend is built with Next.js framework (Vercel, 2025) and Morphic (Miura, 2025), providing a conventional and intuitive user interface that is suitable for petrophysical workflows.

User Interface and Interaction Design. Frontend implementation focuses on user interaction with LAS files, analytical tools and visual outputs. Key frontend features include

- **Drag-and-drop LAS upload** that lets users add files directly in the chat.
- **Real-time visualization** that shows plots and summaries inline including base64-encoded well log images.
- **Chat-based interaction** that lets users request analyses in plain language.

The frontend talks to the backend through defined API routes. An MCP client handles sessions and tool calls so requests remain structured and predictable. The overall system deployment and data processing pipeline are illustrated in Fig. 6.

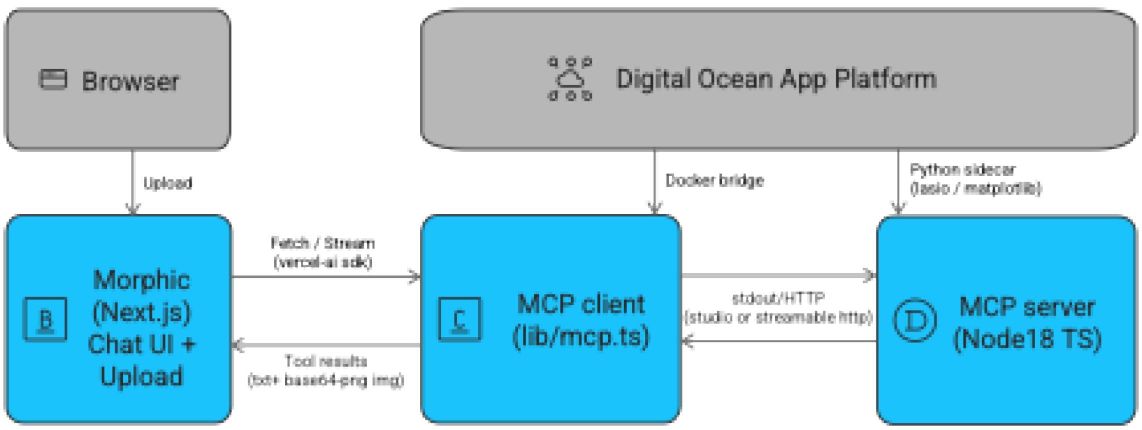

Figure 6—System integration architecture.

The integrated architecture combines a frontend built on Next.js, a dedicated MCP client, and a backend MCP server. User interactions begin in the browser where LAS files are securely uploaded via a drag-and-drop interface. The Morphic frontend communicates directly with the MCP client using fetch/stream protocols. Analytical tasks are orchestrated via structured API requests to the MCP server which executes analytical computations using a Python helper module with the lasio (Inverarity, 2025) and matplotlib libraries. Analytical results, including textual summaries and visual plots in base64 PNG format, are securely transmitted back through the MCP client to the frontend for visualization. The Digital Ocean App Platform (DigitalOcean, 2025) hosts Dockerized backend and frontend components, ensuring seamless container orchestration and reliable data governance.

LLM Integration and Natural Language Orchestration

Integration with large language models such as Gemini 2.5 Pro expands the system's analysis features and keeps the interface easy to use. Natural language orchestration converts plain-text requests into structured MCP API calls so users do not deal with tool details.

Fig. 7 illustrates LLM driven orchestration scheme, detailing the complete execution path of a user's query. A query entered through the Morphic frontend is interpreted by Gemini 2.5 Pro to determine the appropriate analytical operations. These are performed via the MCP protocol and FastAPI backend, which interact with the library to process and visualise petrophysical data. The results are returned to the user interface for real-time display and interpretation.

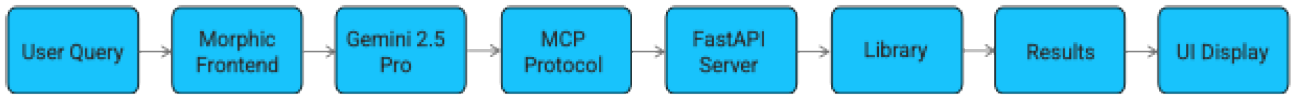

Figure 7—LLM driven workflow orchestration pipeline showing the full lifecycle of a natural language query from user input to result display.

Natural Language Query Processing. The natural language interface uses structured prompts and "chain-of-thought" reasoning techniques to parse and interpret user requests effectively. Queries submitted via the Morphic frontend are processed by the LLM to determine the appropriate sequence of analytical tasks resulting in controlled execution of complex workflows.

LLM Workflow Execution Flow. The implemented LLM driven data workflow follows a sequential process.

- User formulates a natural language query via the frontend.
- The query is relayed to the LLM, in our case Gemini 2.5 Pro, complete with contextual descriptions of available MCP tools and recent session states.
- The LLM identifies relevant MCP tool calls and associated parameters.
- Structured API calls are executed against the MCP server.
- Analytical results, including textual summaries, statistical outputs and plots are returned to the frontend for real-time display and interpretation.

This automated orchestration reduces human error and latency in complex analytical scenarios enabling quick iterative analyses.

MCP Server Implementation (Backend)

The MCP server is the analytical core of the system utilizing Python's FastAPI framework for its asynchronous capabilities and automatic API documentation via Swagger. Fig. 8 illustrates the internal architecture of the MCP server. Incoming HTTP requests are handled by the FastAPI layer which interfaces with the Library MCP Tools module to trigger specific analytical operations via the library. All active sessions are managed in memory using a dedicated Session Manager to provide consistent state management across analytical tasks. Responses are returned to the client after processing enabling secure analysis of well-log data.

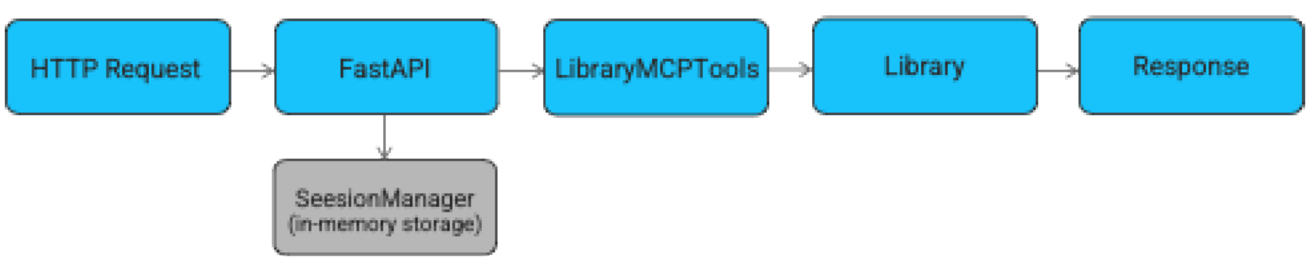

Figure 8—Internal MCP server data flow.

Core MCP Analytical Tools. The MCP server encapsulates 5 key analytical tools, all based on library functionalities, exposing them through standardized, structured MCP APIs.

- **load_las_well** for loading and validation of LAS files. Well log data are parsed into structured in-memory sessions to allow for subsequent analysis.

- **get_curve_stats** that generates detailed statistical summaries (min, max, mean, standard deviation, and key percentiles) for user-specified well-log curves. This assists quick statistical data QC and interpretation tasks.
- **plot_well_log** that creates petrophysical visualizations such as multi-curve well log plots. Generated plots are encoded as base64 PNG images transmitted to frontend clients without intermediate file storage.
- **get_well_info** for retrieving metadata and header information from LAS files to provide contextual understanding of the well's geological and petrophysical attributes.
- **list_curves** enumerates available curves from the loaded LAS file, optionally providing quick statistical summaries for initial screening and validation purposes.

Fig. 9 shows API documentation interface, Swagger UI, for the MCP server. This interface documents the available endpoints for health checks, tool execution, session management and system metrics which highlights the standardized and transparent approach to tool orchestration.

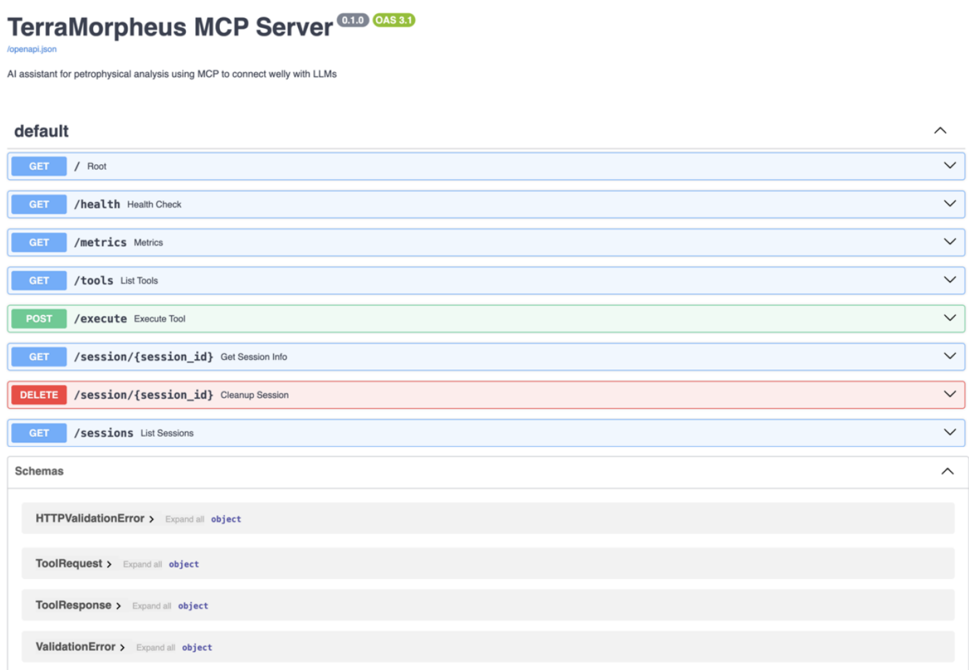

Figure 9—MCP Server API documentation (Swagger UI) showing core endpoints for analytical tool execution and session management.

Session Management and Data Persistence. In the proposed workflow, sessions are processed in memory using a session manager, which ensures that all analytical states are continuously stored throughout the entire interaction with the user and automatically cleared after a configurable 24 hour timeout. The session-based architecture simplifies system administration to comply with corporate security policies.

Testing, Validation, and Continuous Integration

Automated testing procedures are used during the implementation phase; they are necessary to ensure reliability and verify accuracy.

Automated Testing Framework.

- **Module Testing (MCP Tools):** each MCP tool passes thorough isolated testing using pytest, with tests using LAS dataset samples for realistic scenarios.

- **System Integration Testing:** workflow testing verifies complete analytical pipelines from interface interaction to server execution. Docker container-based testing ensures reproducibility and consistency with production deployments.

Test scenarios verify the correctly processing of statistical calculations, accurate visualization, and session consistency when performing analytical tasks.

Continuous Integration and Deployment. Continuous integration workflows using GitHub (GitHub Inc., 2025) automate the validation process, ensuring code quality and deployment readiness across multiple Python versions. Automated container build validations, TypeScript frontend checks, and compatibility assessments further improve system reliability.

Deployment Architecture and Data Governance

Deployment using Docker containers meets corporate security standards. Docker containers are orchestrated through Docker Compose and deployed on Digital Ocean App Platform managed platforms.

- **Containerization**: backend and frontend deployed as independent Docker containers with well-defined communication between containers through internal networks.
- **Security**: environment variables, role-based access control and authentication mechanisms, as well as logging ensure data security.

Technical Implementation Outcomes

The results of implementation demonstrate scalable performance with field data sets. Analytical tasks such as well log data analysis, statistical summaries, and visualization are performed in seconds.

The technical implementation provides a secure analytics platform. It simplifies the petrophysical workflow, improves data transparency, and supports interdisciplinary collaboration solving critical issues in traditional petrophysical data interpretation.

Results and Evaluation

System Performance Evaluation

Performance evaluation of the MCP server with the library shows that core petrophysical workflows such as LAS file import, curve viewing, statistical calculations, and visualization can be efficiently performed on standard corporate workstations. Testing on typical data sets (LAS files ranging from 1 to 10 MB in size) shows that log parsing, data structure initialization, and basic operations are completed in a few seconds. The system architecture is designed without losing session context, so it is possible to work in several stages and also perform iterative analysis if necessary.

Actions are tracked through logging and error handling is provided for file uploads, data verification, and analytical steps. Results, including curve statistics and graphical charts, are generated in standardized formats, that are necessary for quality assurance.

Use Case Demonstrations

The current version includes various application options

- **LAS File Management:** LAS files are loaded and indexed by unique well identifiers (UWI) allowing fast access to data.
- **Curve Quality Control:** available curves can be numbered to obtain static information on them, such as minimum, maximum, average, standard deviation, etc., allowing a quick check of the data.
- **Curve Visualization:** curve graphs are generated and displayed as standard image files, allowing further analysis.

- **Session Context:** loaded responses are stored in the server memory, allowing subsequent operations, such as curve extraction, graphing, or statistical analysis, to be performed without repeated requests.
- **Error Reporting:** for each operation there is a system for detecting errors, such as missing data or invalid files, which are immediately reported to the user to ensure a continuous workflow.

To quantitatively demonstrate the practical impact of the GenAI agent workflow, Table 1 summarizes representative use cases, comparing manual and automated performance, end-user roles, and business/ technical impact.

Table 1—Comparison of manual and agent-based petrophysical workflows with use cases, user roles, efficiency gains and business impact.

Use Case	Workflow Steps (Brief Description)	Time (Manual Workflow)	Time (Agent-Based Workflow)	Primary User Role	Business/ Technical Impact
Batch Quality Control of 50 LAS Files	Upload each LAS file, perform curve header validation	1–2 days	~30 minutes	Petrophysicist	>90% reduction in labor time, consistent QC standards, auditable batch reports, quick onboarding of new datasets
Quick Multi-Curve Visualization	Locate required well(s), extract specified curves (e.g., GR, RES), plot curve tracks, export images for review	15–20 minutes per well	<1 minute per well	Geologist / Petrophysicist	Enables instant screening of multiple wells, allowing faster decision-making in prospect evaluation or well selection
Cross-Disciplinary Data Access and Sharing	Request analytics from petrophysicist, wait for manual extraction and transfer to other teams	1-2 days per request	Immediate(<1 min)	Drilling / Production / Reservoir Engineer	Allows non-specialists to directly access validated data
Iterative Curve QC and Editing for New Data	Sequentially review curve quality, re-run QC after edits, generate updated visualizations after each modification	2–3 hours per dataset	15–20 minutes per dataset	Petrophysicist	Facilitates agile interpretation cycles, speeds up error resolution and QC iterations
Project-Wide Data Screening for Anomalies	Manually browse all wells in a field, flag anomalies, aggregate and summarize outlier statistics for project management	Multiple days (50+ wells)	1–2 hours	Petrophysicist / Team Lead	Empowers data-driven project oversight, ensures early detection of data quality issues

Fig. 10 illustrates a user interaction with the platform inside Cursor IDE. The process starts with a LAS file upload. The AI assistant, based on Gemini 2.5 Pro, parses the user's request, orchestrates the corresponding MCP toolchain and returns structured results. The user interface shows how the system uploads the file, extracts metadata, and performs an initial data quality check, as well as displaying the visualization of several curves and shows expert AI-generated comments in the same chat.

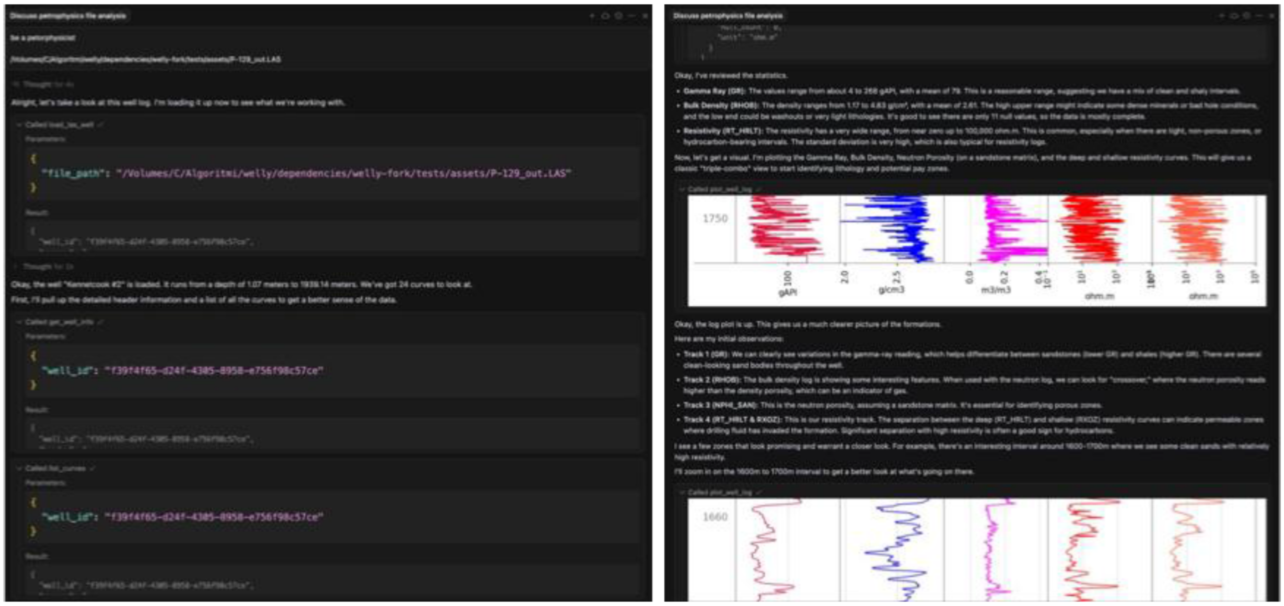

Figure 10—Petrophysical workflow example: sequence of user-AI interactions for LAS file analysis: (a) LAS upload & data preview; (b) Visualization.

Business Impact and Role Transformation

Changes in the Petrophysicist's Role

The proposed architecture involving MCP server transforms the standard workflow of a petrophysicist. Previously a significant amount of time had to be spent to routine data processing and transfer, now the focus shifts to supervising automated processes. Under this approach, the petrophysicist becomes a coordinator responsible for configuring and improving digital workflows. Key responsibilities include developing workflow templates, validating results based on generative artificial intelligence, and adding new modules and custom methodologies as new tasks and data appear. This shift in focus allows petrophysicists to invest more time to high-impact tasks, such as configurating LLM for specific tasks, consulting on complex or non-standard cases, and supporting field operations.

Fig. 11 shows the proposed digital transformation in the petrophysicist's routine work, where the traditional role, characterized by manual compilation, is contrasted with the role using the MCP server and LLM as the main cognitive layer.

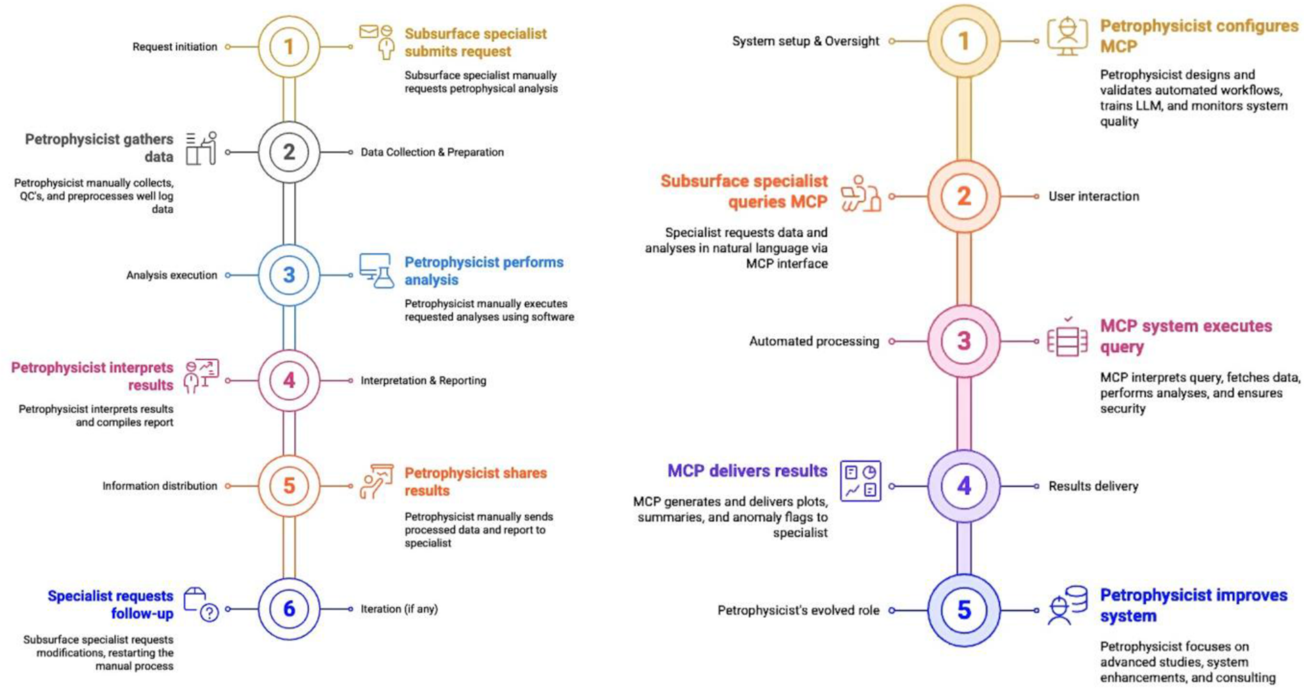

Figure 11—Petrophysical workflow (a) ("As-Is") manual and centralized process; (b) MCP-based.

Workflow Modularity

One of the main advantages of digital architecture using LLM with MCP is its modularity and scalability. The workflow is divided into separate components, each of which is managed via API. This modular approach allows quick adaptation to project requirements and implement new workflows without changing the entire business process. Workflow modules can be independently updated, replaced, and additional modules can be added as well. For example, if new software or a library becomes available, it can be integrated as a separate module without disrupting the entire process.

Another major advantage of this architecture is the ability to connect multiple commercial software such as Petrel, Techlog and other, and enable them to interact via the MCP server.

Future Work and Scalability

The proposed architecture is scalable not only for various petrophysical workflows, but also can be extended to other areas of subsurface development, drilling, surface engineering, and business, supporting the creation of a multi-agent system.

In the future, the authors plan to develop and implement agents for each of the disciplines. One of the challenges is the agent collaboration system and the ability to configure peer-to-peer for direct interaction, based on MCP servers specialized by expertise. Additional verification is also required to allow or restrict access to information depending on the user's role.

As shown in Fig. 12, traditional workflows in subsurface teams and their communication with other teams are characterized by linear request and response cycles through expert collaboration, which often leads to delays in coordination between disciplines. MCP servers like the one presented in this paper, will be developed for different disciplines, and will lead to the proposed multi-agent architecture, presented in Fig. 13, which provides AI orchestration, agent collaboration, and centralized management for improved efficiency.

Figure 12—Traditional data and request flows within subsurface teams.

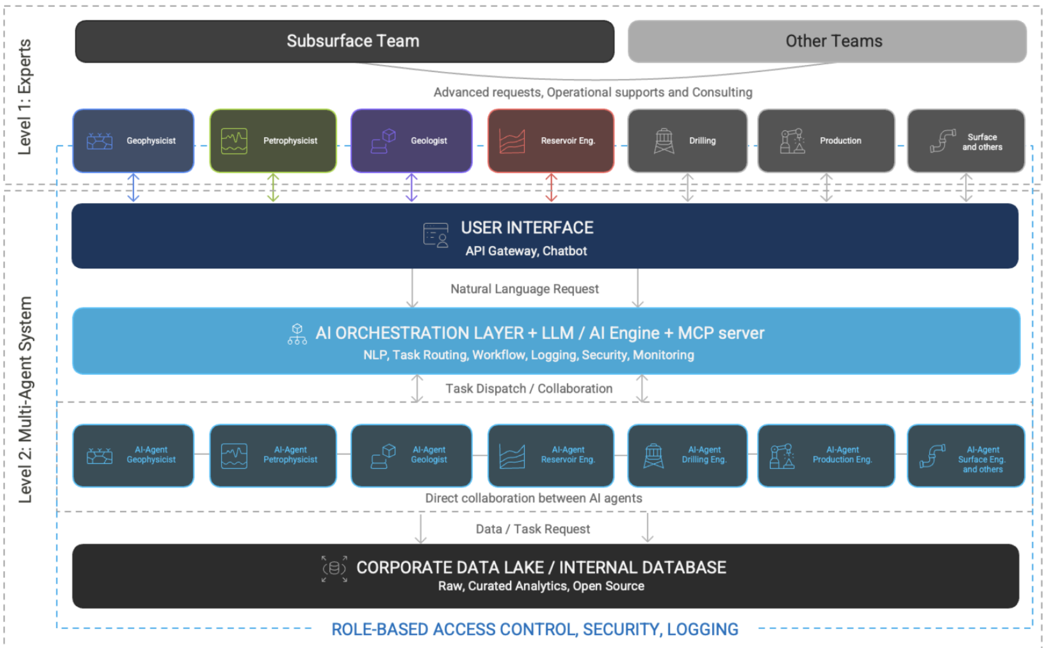

Figure 13—Proposed GenAI-based collaborative architecture: interaction between human experts, AI agents, and internal data infrastructure in subsurface and cross-functional teams.

Conclusion

Summary of Findings

This study demonstrates that the implementation of a modular, digital workflow based on GenAI agent for petrophysical analysis leads to improved operational efficiency, security, and transparency of analysis. A system architecture has been designed and consists of interconnected layers including a Natural Language Interface based on LLM, an MCP server and an open-source Library, which together functioning as a specialized generative artificial intelligence agent for petrophysical analysis.

It is important to note that the redefined role of a petrophysicist is shifting from manual data processing to strategic oversight, control of automated workflows, GenAI agent customisation, and quality assurance. This enables higher-quality analytical results, faster project execution and a digital transformation in subsurface.

Recommendations

Several areas for future development have been identified, such as creation of a multi-agent system, agent-to-agent interaction rules, and access control mechanisms, which could potentially unlock new levels of automation and workflow efficiency within companies. In addition, user experience studies and continuous feedback from interdisciplinary teams will be critical for improving natural language interfaces, ensuring that the system remains intuitive and accessible to both specialists and non-specialists.

Organizations intending to implement similar digital transformation workflow are advised to prioritize secure modular architecture, invest in reliable data governance, and apply a flexible approach to platform customization. To ensure successful adoption and measurable business impact, it is recommended to perform pilot implementations with iterative refinement and experts involvement. Ultimately, the modular agent-oriented framework described in this study serves as a scalable foundation for ongoing digital innovation in subsurface data analysis and cross-disciplinary collaboration.

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